

FULL STACK DEVELOPMENT ASSIGNMENT - 1

AIM: develop responsive web designs using HTML5, containing a form. Style the pages using CSS. Use of tag selector, class selector and id selectors. Use Inline, Internal and External CSS. Apply Bootstrap CSS.

OBJECTIVES: i) to understand HTML tags
ii) to learn styling of web pages using CSS
iii) to learn Bootstrap Front End Framework

THEORY:

i) Responsive Web Design: (RWD)

- It is an approach to web design that makes web pages ~~that~~ render well on a variety of devices and window or screen sizes. This layout adapts automatically to the screen size, orientation and platform, ensuring usability and aesthetics across desktop, tablets and smartphones.
- Primary Goal: provide an optimal viewing and interaction experience - easy reading and navigation with minimal resizing, panning and scrolling - across wide range of devices

ii) Role of `<meta name="viewport">`

- `<meta name="viewport">` ~~control~~ tag tells the browser how to control the page dimensions and Scaling
- Role: it sets the visible area of a web page to match the device's screen width and initial zoom level
- Essential for RWD: without this tag, mobile browsers assume a default viewport width, which causes the page to appear zoomed out and not scale to small screens

iii) Bootstrap & Grid System.

- bootstrap helps create responsive website that adapt to different devices (mobile, table, desktop)
- uses a 12-column grid system where content is placed inside row and col elements.
- the grid adapts using breakpoints with prefixes:
 - col- → extra small (< 576px)
 - col-sm → small (≥ 576px)
 - col-md → medium (≥ 768px)
 - col-lg → large (≥ 992px)
 - col-xl → Extra large (≥ 1200px)

iv) Difference between Tag, Class & ID selectors

Selectors	Symbol	Purpose	Example
→ Tag	none	styles all <p> tags	p { color: red }
→ Class	.	styles all elements with given class	.highlight { background: yellow; }
→ ID	#	styles one specific element with an id	#main { font-size: 20px }

v) Three Ways to Apply CSS

i) Inline CSS: written directly in HTML element's style attribute

```
<p style = 'color: blue;' > Hello </p>
```

ii) Internal CSS: written inside a <style> tag in <head>

```
<style>
```

```
p { color: green; }
```

```
</style>
```

iii) External CSS: stored in a .css file and linked in <head>

```
<link rel = 'stylesheet' href = 'style.css' >
```


PROBLEM STATEMENT:

F) Personal Portfolio Website (Roll no. 13 to 24)

CONCLUSION:

In this assignment, I learned how to build responsive web pages using HTML5, CSS and Bootstrap. I understood the use of different CSS selectors, the ways to apply CSS, and how the viewport tag with Bootstrap's grid system ensure layouts adapt to all screen sizes.

31
12/8/25.

ASSIGNMENT-2

WEB TECHNOLOGIES LABORATORY

AIM: Develop a web application using javascript to implement sessions cookies, DOM. Perform validations such as checking for emptiness, only numbers for phone number, special character requirement for password, regular expression for certain format of the fields etc. Use the my SQL database.

OBJECTIVES:

- i) to understand what form validation is
- ii) to learn basic functioning of DOM objects
- iii) to learn how to apply various techniques to implement it.

THEORY:

i) Role of Regular Expression in Validation

- Regular expressions (regex) are patterns used to match strings
- They are suitable for validating forms because they allow concise definitions of rules:
 - `/^[0-9]{10}$/` ensures a phone number is only digits & 10 digits long
 - `/^(?=.*[A-Z])(?=.*[0-9])(?=.*[&$#@]).{7}$/` ensures a strong password
- Regex reduces complex string checking logic into compact and efficient expressions

ii) Difference b/w Session and Cookie

Session	Cookie
- Storage: stored on server - more secure - lifetime ends when browser is closed - Tracks user state (eg: logged-in user info)	- Storage: stored on client browser - less secure - lifetime: can be set with expiry date - stores user preferences (eg: theme number etc)

How they work together:

- when you log in, the server creates a session and stores details
- a cookie stores the session ID in the browser
- On each request, the cookie passes the session-ID -> server matches session -> maintain login state

iii) Purpose of Client-side & Server-side Validation

→ Client-side (JavaScript / DOM validation)

- immediate feedback to user
- reduces server load
- example: Showing 'Phone must be 10 digits' instantly.

→ Server-side (PHP, Python, Node, Java etc + MySQL)

- final line of defense
- Protects against malicious input or bypassing of client-side scripts

A hacker disables JavaScript in the browser and submits invalid/malicious SQL code in the form -> leads to SQL Injection attack.

iv) Provide a simple example of how Javascript can interact with the DOM to dynamically change the content of a web page after a user action, such as a form submission.

```
→ <P id = "message"> Hello! </P>  
< button on click = change text () "> click me  
</button>  
<Script>  
function change text () {  
document get Element By ID ("message") inner  
HTML <= "you" }  
</Script>
```

v) Give the steps for connectivity from front end using HTML CSS JS to mysql

- i) front end: create form in HTML and collect data
- ii) Java script: optionally, validate before submission
- iii) Send data: Use AJAX / Fetch to send data to backend
- iv) Backend: Receive data, validate again then connect to MySQL
- v) MySQL: use a data base driver / library to insert / query data
- vi) Response: backend sends a success / failure response back to front end

CONCLUSION: understanding key concepts such as regular expressions, sessions & cookies, form validation, DOM manipulation for building secure efficient & user friendly applications.

FAQ's

i) Write 3 reasons why form validations are important.

- i) prevents submission of invalid/incomplete data
- ii) improves user experience with instant feedback
- iii) protects server and database from malicious input

ii) Give an example of how to modify an attribute value using DOM:

eg) document.getElementById('My Input').setAttribute('Placeholder', 'Enter your name')

iii) What are different features of Javascript?

- > client side scripting language
- > event driven and asynchronous
- > supports OOP, functional programming
- > manipulates DOM
- > supports AJAX for asynchronous requests
- > supports prototypal inheritance

27
2/9/25

ASSIGNMENT 3

AIM: Design an interactive front end application using React by implementing templating using components, state and Props, Class, Events. It must be responsive to scale across different platforms.

OBJECTIVES: To develop a responsive, interactive front end application using React.js that effectively demonstrates the fundamental concepts of component-based architecture, state management and event handling. The application will serve as a practical exercise in building a scalable user interface by implementing templating with components, managing dynamic data with states and props, and handling user interactions with events, ensuring a seamless user experience across various devices and screen sizes.

THEORY:

i) Explain Role of the state and props in React.

Ans. **State:**

- managed inside a component
- mutable: can be updated with `setState` or `useState`
- used to store dynamic data that changes over time (like form input, counters, toggles)

Props:

- passed from parent to child components
- immutable: cannot be modified by the child

- used for data sharing & configuration
ex: a button label passed from a parent
props: external, controlled by parent, read-only
state: internal, controlled by component, write available to edit
- props: enable component reusability & data flow
- state: manage interactive dynamic behaviours within in a component.

2. What is React component? Diff. b/w a class component functional component.

Ans A react component is a reusable, independent piece of UI, like card or form

- Class components:
 - written as ES6 classes
 - uses `render()` method to return JSX
 - can use the state & lifecycle methods
- functional components:
 - written as plain functions, returning JSX
 - use hooks like `useState`, `useEffect` for state & side effects

Advantages of Functional Components with hooks:

- ~~the~~ simpler, short syntax
- no need to worry about this keyword
- hooks like `useState`, `useEffect`, `useContext` give access to state & life cycle in a clean way
- easier to test & optimize, has smaller reusable logic with custom hooks.

3. Describe the concept of 'templating using components' in React. Why is this approach considered superior to traditional web development methods that rely on monolithic HTML files?

Ans - Instead of one giant HTML file, react breaks the UI into small, reusable components.

- each component manages its own logic & style

Why better than traditional HTML:

- Reusability: write once, use everywhere

- maintainability: easy to update just one component

- dynamic data: react renders data driven UI
Static HTML

4. How do you handle user events in React?

Ans Events are handled with camel case & by passing functions not strings.

Ex:

```
import { useState } from 'react'
```

```
function clickExample () {
```

```
  const [message, set message] = useState ('Hello'); }
```

```
  function handle click () {
```

```
    set message ('You clicked the button'); }
```

```
  return (
```

```
    <div>
```

```
      <p> {message} </p>
```

```
      <button onclick = { handle click } > click me </button>
```

```
    </div> ); }
```

event is attached on click, function updates

state → react re-renders updates automatically.

5 Responsive design makes a website adapt to different devices (desktop, tablet, mobile)

It ensures usability accessibility & better user experience

CSS media queries: adjust layout based on size

CSS in JS: write responsive styles directly in components

Ex:

- container { width: 80%; }

@ media (max-width 768 px { container } width: 100%; }

CONCLUSION:

We learnt to develop a responsive interactive front end application using React.js by implementing templating using components, states and props, class, Events.

23/9/25

ASSIGNMENT- 4

AIM: Enhance web page developed in earlier assignment by rendering lists and Portals, Error handling, Routing and style with React CSS also make it a responsive design to scale well across PC, table and mobile phone

OBJECTIVE: Enhance User Interface & Experience
improve application robustness & navigation

THEORY:

i) How do lists & Keys work in React?

Ans) Lists in React:

- In React, you can render lists of data by using the `map()` method on array.
- Each item in the list gets rendered as a React element

Keys in React:

- special string attribute used to uniquely identify elements in a list or an array of components. Key help React efficiently update & re-render only the necessary components when the state changes
- when you render a list of items dynamically, React needs to know which items have changed, been added or removed

ii) What is a react portal & when would you use one?

Ans) React Portal:

- A portal is a feature in React that allows you to render a child component to be rendered in different parts of the DOM, which is particularly useful for

things like modals, tooltips, or drop-down menus where you might need them to render outside the normal flow of the component tree.

when to use:

use a portal when you need to render a child component outside the current DOM hierarchy, typically for overlays, modals or tooltips where the component needs to appear outside the usual flow of the parent.

iii) Discuss the importance of Error Boundaries in React.

Error Boundaries:

- Error Boundaries are react components that catch JavaScript in their child component tree, log those errors & display a fallback UI, instead of crashing the whole app. Without error boundaries if an error occurs in any part of the app, it could unmount the entire component tree.

They are important because:

- Prevent ~~the~~ crashes: without them a single child component could bring down the entire app.
- Graceful degradation: fallback UI allows users to still navigate & run basic functions.
- Error logging: Error boundaries allow you to log errors to external services like Sentry or LogRocket to improve the app's stability over time.

iv) How does react router enable single page application (SPA) functionality?

React router allows navigation between pages without refreshing the browser. It turns a react app into a single page application (SPA) by dynamically updating the view when the URL changes.

Benefits:

- faster navigation
- smooth user experience
- route based rendering

5. Explain the different ways to style a react application.

- a) CSS style sheets: traditional CSS files imported into components
- b) Inline styles: directly style elements using JavaScript objects
- c) CSS modules: scoped CSS that avoids naming conflicts.
- d) CSS in JS libraries: styled component, emotion - styles written in JS
- e) UI Frameworks: Bootstrap material - UI, tailwind CSS for pre-built responsive designs.

CONCLUSION:

We learnt how to enhance user interface & experience and improve application robustness & navigation.

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28/9/25

ASSIGNMENT - 5

AIM: develop a responsive web design using express framework to perform CRUD operations and deploy with Node Js use Mongo DB

OBJECTIVES: Develop a full stack application demonstrate backend development & ~~development~~ deployment proficiency.

THEORY:

i) What is the role of express Js as a web framework for node Js?

Ans) Express Js is a minimal and flexible web framework built on top of node Js. It provides powerful ~~basic~~ features such as routing middleware support, request & response handling and template rendering which makes backend development faster & easier instead of writing complex server code with just node js, developers use express Js to build APIs, full stack applications & scalable web servers with less efforts & more structures.

ii) Explain the concept of CRUD operations in the context of a web application.

C : create → add / create new data / database

R : read → retrieve / read existing data

U : Update → modify / update existing data / values

D : delete → delete / remove data

iii) Why is MongoDB a suitable choice for this project?
Ans) MongoDB is a document oriented NoSQL database that stores different types of data without needing a fixed structure which speeds development. MongoDB also supports scalability, high performance & integration with Express.js. Through libraries like Mongoose it is a perfect choice for modern web applications, where data requirements may change frequently.

iv) What steps are involved in deploying a Node.js and Express application?

Ans) Deploying a Node.js & Express application involves the following steps:

- i) develop locally: build your app with Express routes & MongoDB integration
- ii) version control: push your code into Github or another repository
- iii) Choose hosting: use platforms like Heroku, Render or Digital Ocean
- iv) Install dependencies: run `npm install` on the server
- v) Configure environment variables: Set up MongoDB connection ports etc.
- vi) Run APP: Start using `node server.js` or a process manager like PM2
- vii) Test Deployment: ensure routes & DB connections work correctly online

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